32-cell monitor interface

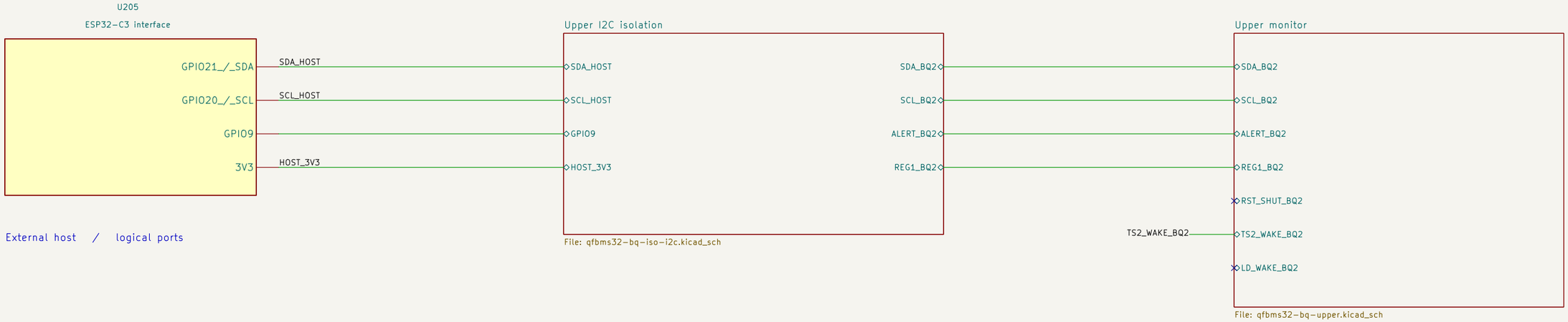
Charge-only BMS / 100 kHz I2C / ESP32-C3 host

HOST DOMAIN / CELL0

UPPER DOMAIN / CELL16

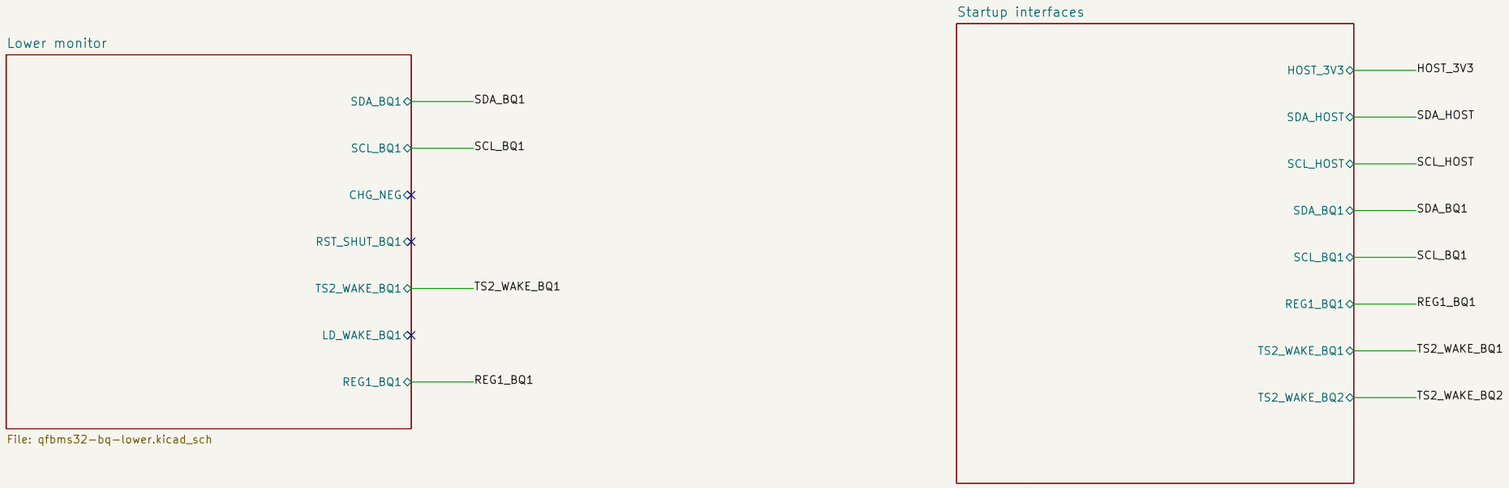
U204 TMUX154E > U_ISO ISO1640

Domain boundary inside ISO1640



External host / logical ports

Cells 17-32 | upper address 0x08



Cells 1-16 | lower address 0x08 at reset

Power and enumeration

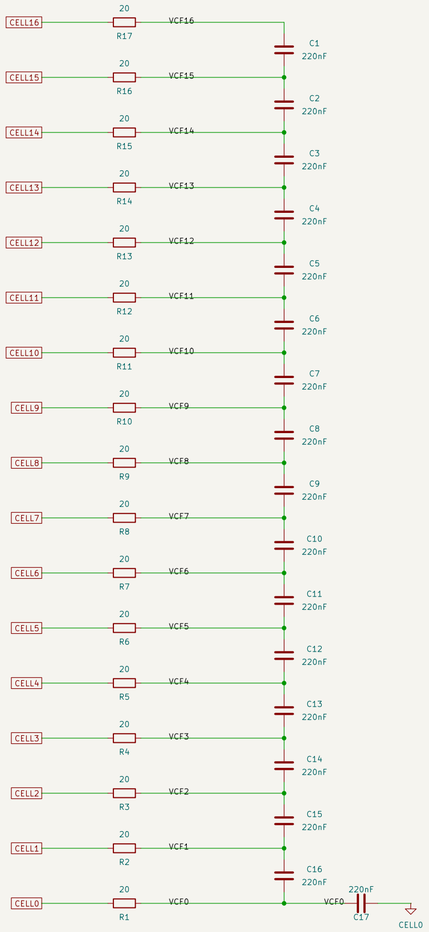
1. Start host power. Pulse WAKE.
2. Wait for both local REG1 rails.
3. Keep GPIO9 high; configure lower.
4. Verify lower at 0x10.
5. Drive GPIO9 low; admit upper at 0x08.

Schematic increment only. qfbms32.kicad_pcb is not updated from this file.

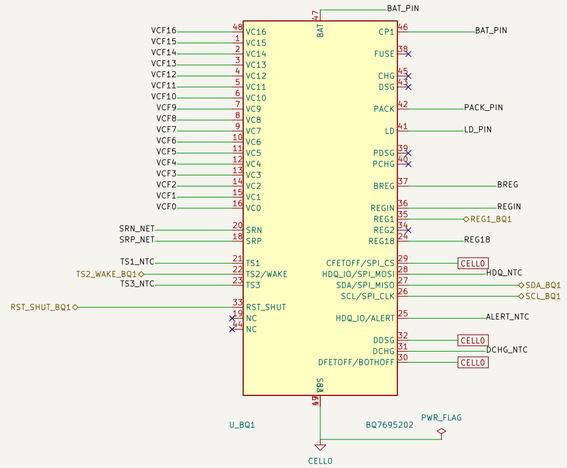
Sheet: /		
File: qfbms32-bq-monitors.kicad_sch		
Title: QFBMS32 BQ monitor sheets		
Size: A3	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 1/5

Lower monitor / cells 1 to 16 / reference CELL0

01 Cell input filters

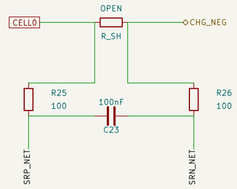


02 Monitor pinout



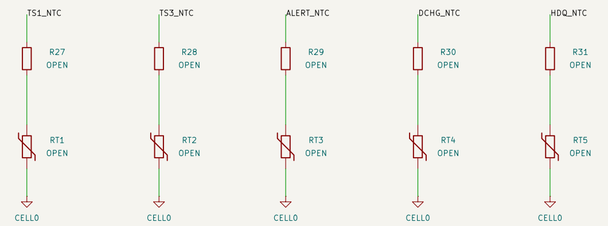
Lower SDA/SCL reach the host via U206.
Host power sequencing and bus thresholds are unresolved.

03 Current sense



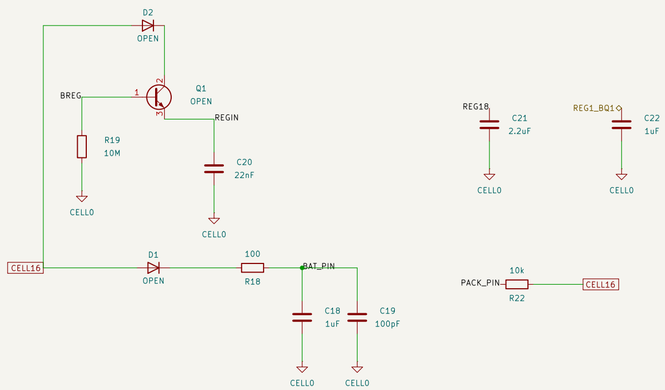
OPEN Confirm Kelvin pickup, shunt value and firmware sign/gain.

04 Temperature inputs



OPEN Firmware assumes 10 kOhm / beta 3380 and 500 ohm series correction.

05 BAT and local regulators



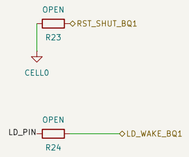
OPEN Select stack-rated diodes, REGO transistor and passive ratings. Check REGIN headroom and dissipation.

06 I2C pull-ups



OPEN Select pulls at 100 kHz. Check rail limits, sink current and rise time for this segment.

07 Wake interface



TS2 contact returns to CELL0.
Pull low, then release. REG1 can be off.
Drivers: Startup interfaces. Hot leakage OPEN.

Charge-only monitor sheet. Do not import this netlist into qfbms32.kicad_pcb.

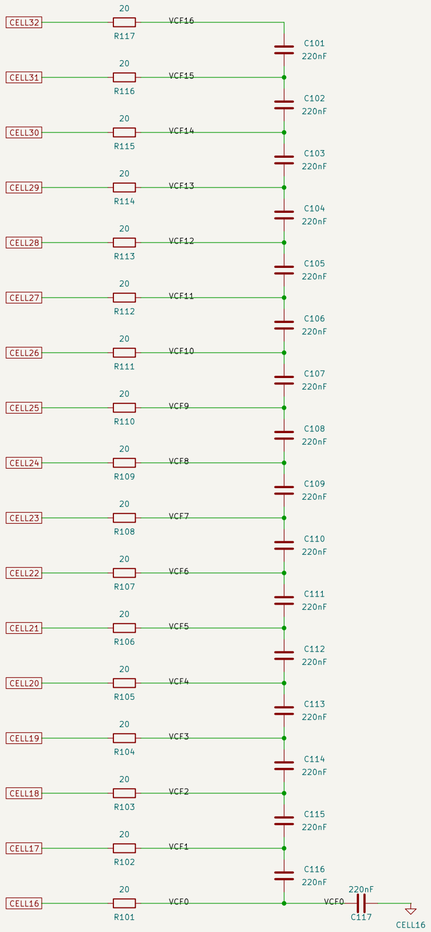
Sheet: /Lower monitor/
File: qfbms32-bq-lower.kicad_sch

Title: QFBMS32 lower monitor U_BQ1

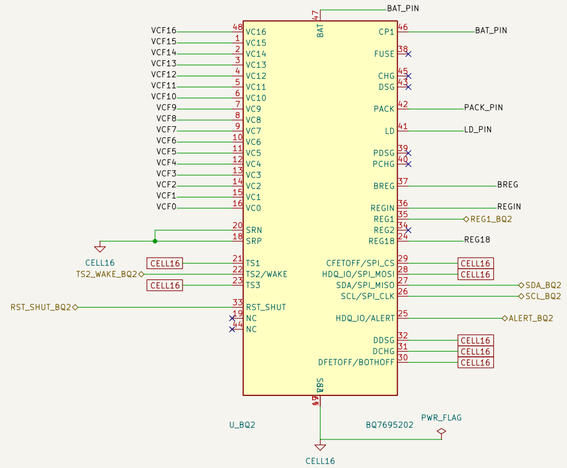
Size: A2	Date: 2026-11-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 2/5

Upper monitor / cells 17 to 32 / reference CELL16

01 Cell input filters



02 Monitor pinout



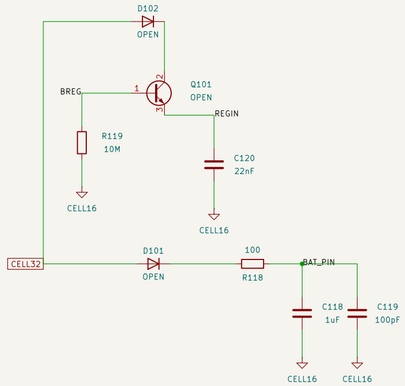
03 Current sense

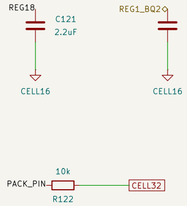
SRP (18) and SRN (20) share upper VSS (17).
The current shunt is on the lower monitor.

04 Temperature inputs

Internal temperature only. Unused sensor inputs return to CELL16.
ALERT terminates on the interface sheet.

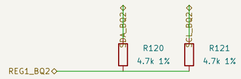
05 BAT and local regulators





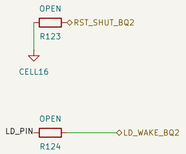
OPEN Select stack-rated diodes, REGO transistor and passive ratings. Check REGIN headroom and dissipation.

06 I2C pull-ups



OPEN Select pulls at 100 kHz. Check rail limits, sink current and rise time for this segment.

07 Wake interface



TS2 contact returns to CELL16.
Pull low, then release. REG1 can be off.
Drivers: Startup interfaces. Hot leakage OPEN.

Charge-only monitor sheet. Do not import this netlist into qfbms32.kicad_pcb.

Sheet: /Upper monitor/
File: qfbms32-bq-upper.kicad_sch

Title: QFBMS32 upper monitor U_BQ2

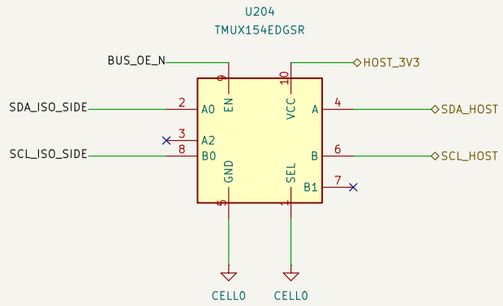
Size: A2	Date: 2026-11-04	Rev: 0.2
KiCad E.D.A. 10.0.6		Id: 3/5

Upper monitor communication

CELL0 host domain

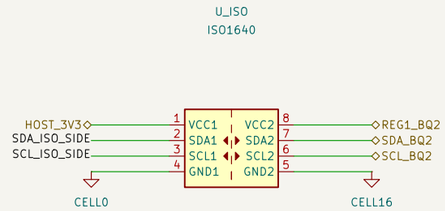
CELL16 upper domain

01 GPIO9-controlled disconnect



GPIO9 high opens both lines. Low selects A0/B0 after the lower address check.

02 ISO1640 domain boundary



OPEN Select ISO1640 orderable 8-D suffix. Check rail margins and sequencing.

03 Host pull-ups and reset bias



OPEN R201/R202: 4.7 kOhm candidate at 100 kHz. Verify rise time, sink current and 80 pF side-1 limit.
R203 biases OE high through reset. Install R204 for normal operation.



Remove R204 before strapping GPIO9 low for download. R203 then holds U204 open.

Upper ALERT stops here; host firmware does not read it.

04 Local bypass

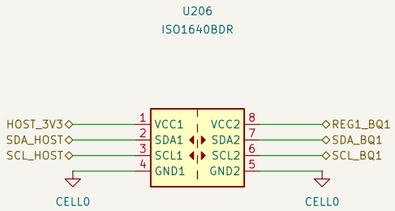


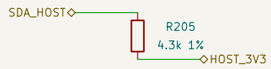
GPIO9 is pulled high at reset. GPIO9 low selects ESP32-C3 download mode.

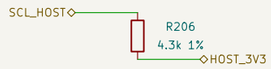
Sheet: /Upper I2C isolation/		
File: qfbms32-bq-iso-i2c.kicad_sch		
Title: QFBMS32 upper I2C isolation		
Size: A3	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 4/5

Host startup interfaces

01 Lower I2C / shared CELL0 return



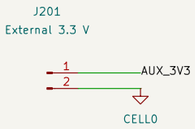








02 Bench host power

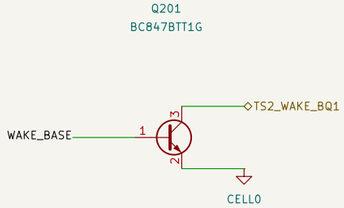




Close SW201 to start the host with both REG1 rails off.
Production auxiliary supply, protection and hold-up OPEN.

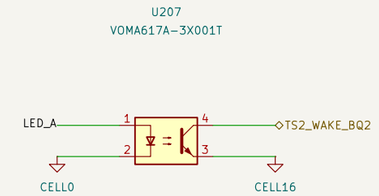
03 Manual wake / host power must be valid













SW202 owns both wake paths. Release before requesting SHUTDOWN.
No ESP output drives the base or LED. Hot released leakage OPEN.

Bench interfaces only. Production host power and automatic wake remain OPEN.

Sheet: /Startup interfaces/
File: qfbms32-bq-startup.kicad_sch

Title: Host startup, lower I2C and dual wake

Size: A3	Date: 2026-11-04	**Rev: 0.2**
KiCad E.D.A. 10.0.6		Id: 5/5